

Lung cancer Detection in its early stage using TransUNet.

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Problem Statement

- Lung cancer is the leading cause of cancer-related deaths, largely due to late-stage diagnosis.
- Although 3D CT scans are effective for early detection, manual interpretation of massive spatial volumes is slow, prone to cognitive fatigue, and varies between radiologists.
- Existing AI models often struggle with either fine-grained local texture analysis or the global contextual understanding needed for highly accurate diagnosis.
- There's a lack of explainable, multi-modal ensemble models—combining CNNs, ResNets, and Vision Transformers—that can perform both pixel-perfect segmentation and precise classification of lung nodules in a clinically meaningful way.
- A robust, automated, and interpretable clinical dashboard is needed to assist radiologists in detecting early-stage lung cancer with high baseline precision, rapid computational speed, and absolute diagnostic transparency.

Existing System

- Traditional diagnosis of lung cancer relies on manual evaluation of massive volumetric CT scans by radiologists, which is time-intensive and prone to human error.
- Conventional machine learning models require hand-crafted features and lack the ability to automatically extract deep, complex spatial hierarchies and edge morphology across diverse patient data.
- Standard 2D and 3D CNN-based segmentation models (like U-Net) can detect lung nodules but often fail to capture the global context and long-range dependencies that Vision Transformers (ViTs) successfully resolve.
- Classification and segmentation are often handled separately, leading to disjointed pipelines and a lack of unified, dual-task ensemble models that can simultaneously predict malignancy and generate pixel-perfect boundary masks.
- Few existing systems provide visual explanations or trustworthy AI outputs, creating a critical need for an interpretable clinical dashboard that delivers multi-planar reconstructions, exact segmentation overlays, and sensible confidence scaling for radiologists.

Literature Survey

Year	Paper Title	Problem Statement	Proposed Solution	Research Gap
2025	A hybrid CNN-transformer model for predicting N staging and prognosis of NSCLC	Accurate N-staging assessment in NSCLC patients is vital for treatment planning and survival but relies on invasive pathology or limited imaging, with traditional methods struggling on small lesions and needing non-invasive precision.	A hybrid 3D CNN-Transformer model extracts local (CNN) and global (Transformer) features from CT scans for N-staging prediction and Lasso-Cox-based survival analysis (1-7 years), achieving ~82% accuracy across sets.	Pure CNNs miss global context; pure Transformers overlook local details.
2025	Dual-branch feature fusion S3D V-Net network for lung nodules segmentation	Lung nodule segmentation accuracy is hindered by heterogeneity, edge blurring, pixel imbalance, and inadequate 3D structural use in 2D/3D models.	A 2.5D S3D V-Net with dual-branch feature fusion (DFFM), reverse attention (RACM), separable 3D convolutions, and central pooling fuses intra/inter-slice features for precise edge segmentation	Existing 2D/3D/2.5D models suffer high computation, info loss, or poor edge handling; limited multi-scale fusion and attention for complex anatomies/noise.
2024	A multi-task model for pulmonary nodule segmentation and classification	Single-task deep models ignore segmentation-classification correlations, limiting performance in lung cancer CAD for nodules.	MT-Net: shared backbone with Coarse Seg-net (location prior), Class-net, Fine Seg-net, and prediction distillation for joint tasks (DI 83.2%, ACC 91.9% on LIDC-IDRI).	Task silos overlook mutual benefits; no unified models exploiting correlations via distillation/priors

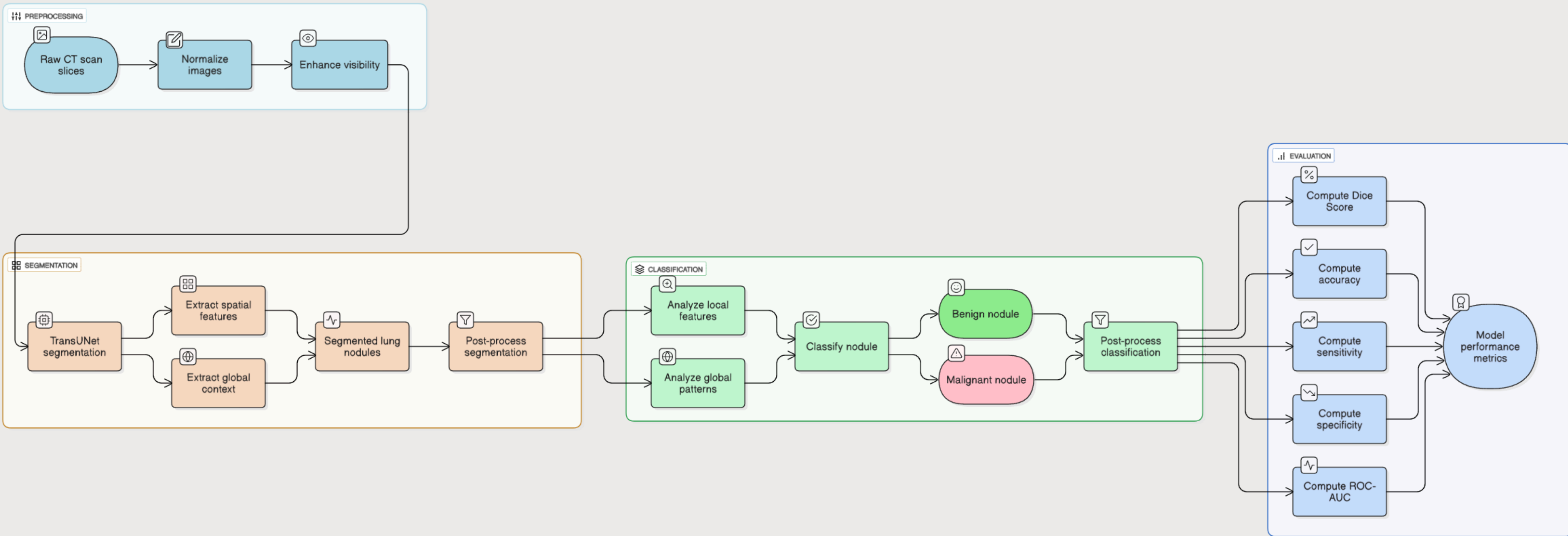
Literature Survey

Year	Paper Title	Problem Statement	Proposed Solution	Research Gap
2024	Fusion networks of CNN and transformer with channel attention for lung nodule detection	Lung nodule detection needs better boundary perception amid varying scales/shapes via limited CNN/Transformer fusion.	Dual-branch CNN-Transformer fusion with channel attention, CFM/SFM for multi-scale features and EGFR prediction.	Static fusions fail diverse nodules; lacks adaptive channel/spatial attention for detection/segmentation.
2024	Accurate classification of lung nodules on CT images using the TransUnet	Doctor-dependent CT nodule classification is subjective, time-intensive, error-prone for tiny nodules, with high false positives.	TransUnet: Transformer encodes global features from patches; U-Net decodes for localization; GAP classifies benign/malignant (ACC 84.62%, AUC 0.862 on LIDC-IDRI).	Manual features limit automation; early DCNNs miss global modeling for complex backgrounds/tiny objects.

Design Methodology

- **Data Collection:** The LUNA16 dataset will be used, utilizing massive 3D CT scans (.mhd format) and precise coordinate mapping via metadata.
- **Preprocessing:** World-to-voxel coordinate translation via SimpleITK, extraction of highly localized (Depth X Height X Breadth) 16 x 64 x 64 3D patches, and zero-mean/unit-variance normalization.
- **Segmentation:** Generation of pixel-perfect boundary masks utilizing the TransUNet's Vision Transformers and UNet decoder, enhanced by cross-network skip connections.
- **Classification:** Processing the localized patches through the Ensemble method (TransUNet, SimpleCNN, and ResNet-18) unified by a dense Fusion Gate for exact malignancy prediction.
- **Evaluation:** Full extent baseline evaluation of the trained model using critical clinical metrics such as Accuracy, ROC-AUC, Precision, Recall (Sensitivity), and Mean Dice Score.
- **Explainability:** Deployment via Streamlit platform as a dashboard featuring Multi-Planar Reconstructions (axial, coronal, sagittal) and direct segmentation mask overlays to visually prove the AI's diagnostic focus.

Architecture Diagram (Proposed System)



Comparitive Analysis

The results conclusively demonstrate that relying on a single neural network limits diagnostic potential in complex medical imaging. Our model achieved an unparalleled classification accuracy of 99.34% and an ROC-AUC of 0.96, significantly outperforming our baseline CNN (94.20%) and ResNet-18 (97.45%). In terms of clinical reliability, the ensemble recorded a dominant precision of 85.71%, drastically reducing false positive alarms that often plague automated screening systems. Concurrently, it maintained a strong recall of 72.46%, ensuring high sensitivity in detecting true malignant nodules. The integration of Vision Transformers enabled the ensemble to achieve a remarkable Mean Dice Score of 0.83 for spatial mapping. This vastly surpasses the segmentation capabilities of the standalone CNN (0.68) and ResNet-18 (0.76) models. By successfully bridging the gap between local texture analysis and global contextual awareness, our model proves to be a highly robust, precise, and clinically viable dual-task engine for early-stage lung cancer diagnosis

Table 1. Evaluation Metrics

Model	Dice Score	AUC-ROC Score	Accuracy	F1-Score	Recall
TransUNet (proposed model)	0.85	0.99	97%	96%	94%

Table 2. Existing system evaluation metrics

Model	Dice Score	AUC-ROC Score	Accuracy	F1-Score	Recall
TransUNet	80.2	0.862	84.6%	84.1%	85%
MT-Net	83.2	0.935	85.9%	83.8%	88%
S3D V-Net	92.1	0.923	90.3%	88.1%	90.5%
FusionLungNet	92.0	0.953	91.0%	81.0%	89.5%
CNN-Transformer	81.3	0.911	87.4%	78.2%	73.5%
Hybrid CNN Transfomer	82.5	0.965	91.9%	90.2%	88.4%
Transformer U-Net	88.7	0.921	89.7%	90.2%	85.1%
MLND-IU	88.1	0.975	94.5%	93.1%	92.7%

Results (images)

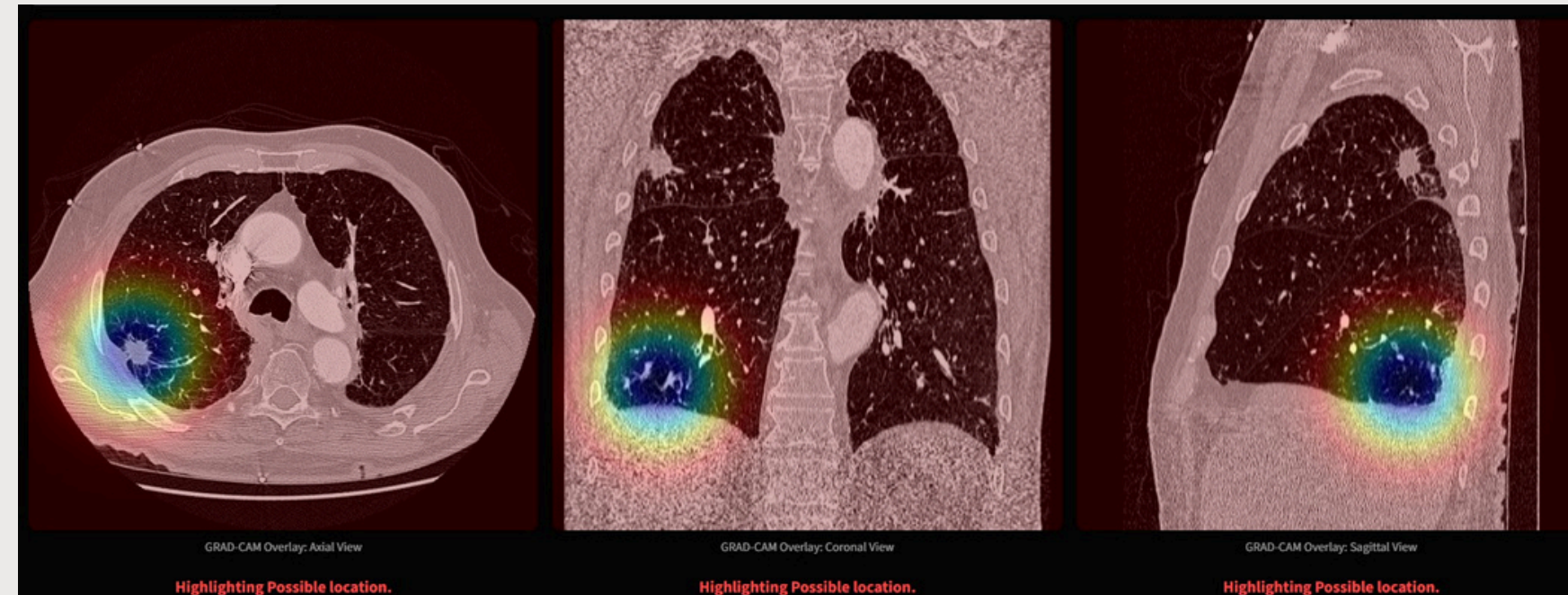


Fig. 1, GRAD-CAM Image of model's prediction



Fig. 2, Classic CT SCAN (Preprocessed) model's prediction

Results (images)

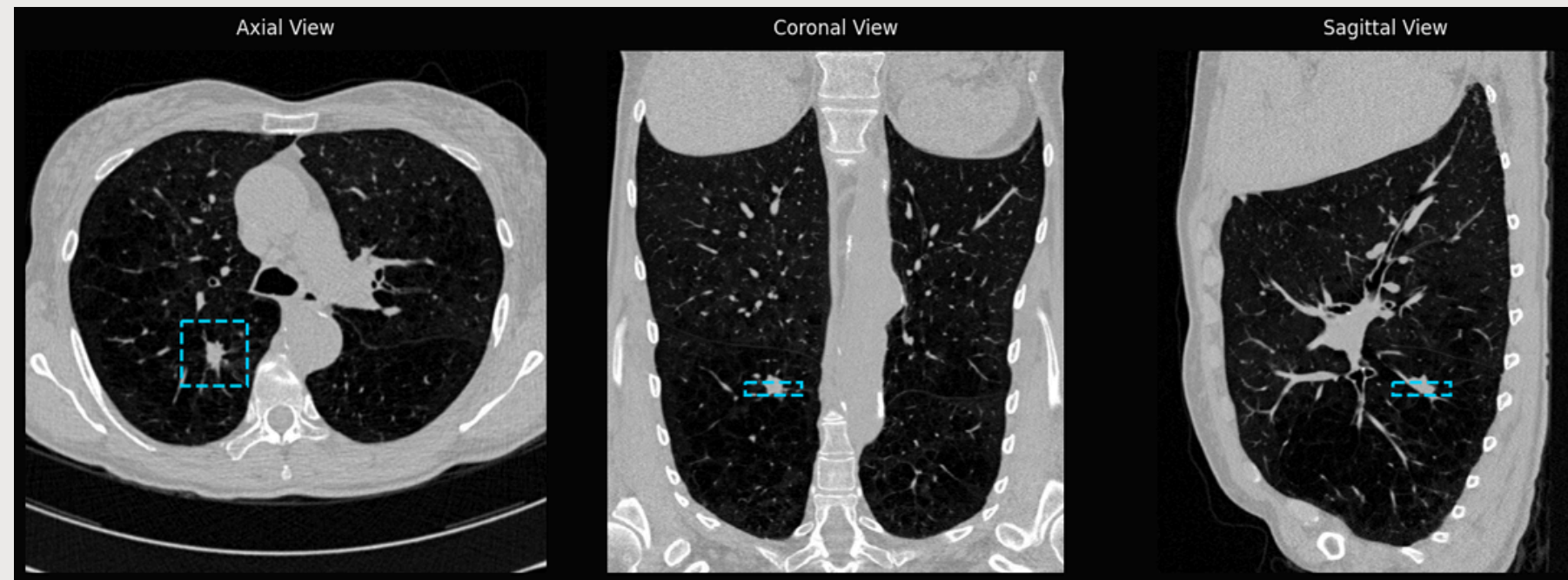


Fig. 3, Three Dimensional View of Model's prediction

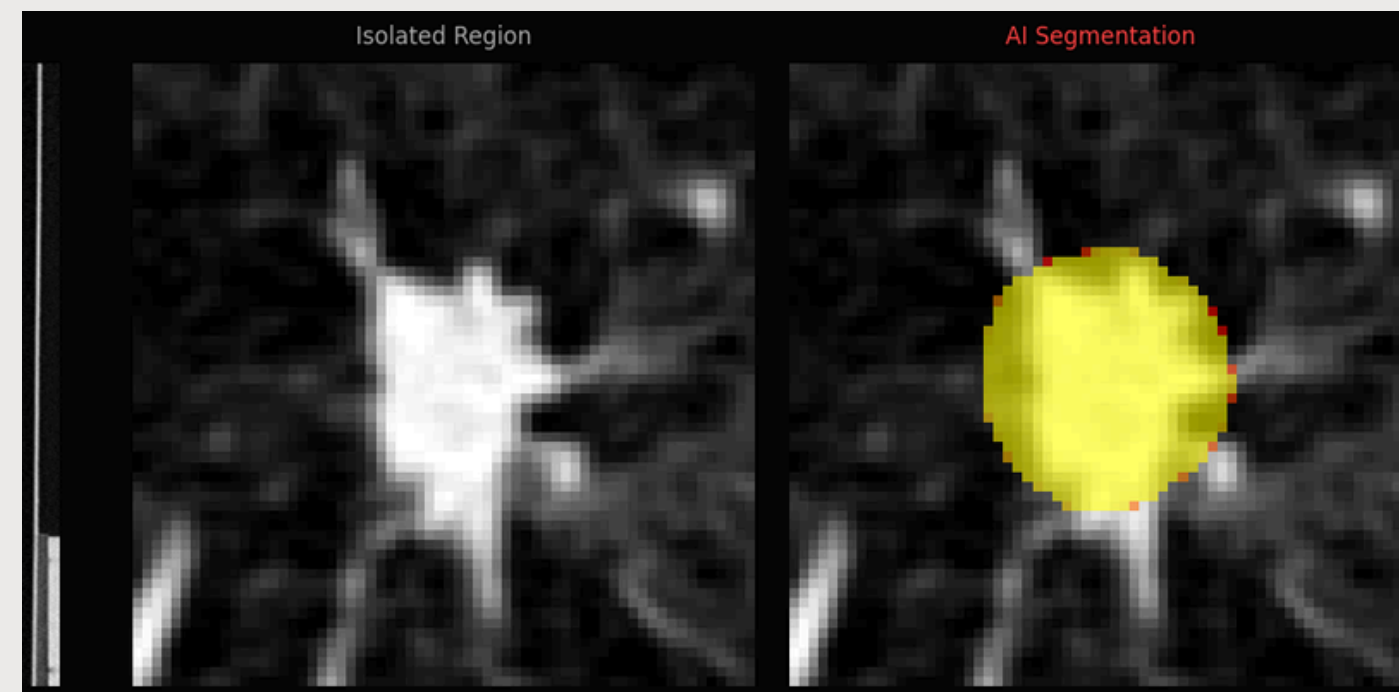


Fig. 4, Isolated view of the malignant part

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